

Teen Numbers — Fixing Swapped Tens and Ones

Answer Key

Kindergarten–Grade 1

Quick Answers

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|-------|-------|-------|
| 1. 14 | 3. 18 | 5. 17 |
| 2. 16 | 4. — | 6. — |
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Common wrong answers

1. *If they got 41: counted the ones as tens and the ten as ones (swapped)*
 2. *If they got 61: wrote the ones digit in the tens place (swapped)*
 3. *If they got 81: wrote eight-teen as 8 tens and 1 one (swapped)*
 5. *If they got 71: read the full ten-frame as ones and the loose dots as tens (swapped)*
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1. One full ten-frame and 4 loose dots. Write the number.

Step 1: the full ten-frame is worth ten, so there is 1 ten. Step 2: the loose dots are the ones, so there are 4 ones. Step 3: the tens digit is written first, the ones digit second: $10 + 4$. The number is 14.

Watch for: a child who writes 41 has read the loose dots as tens.

2. Leo builds 1 ten-stick and 6 ones and writes 61. Write the number his blocks really show.

Step 1: name the parts — 1 ten-stick means 1 ten, 6 loose cubes means 6 ones. Step 2: the tens digit goes in the first (left) place and the ones digit in the second place, so $10 + 6$ is written with a 1 first and a 6 second. Step 3: Leo wrote his 6 first, which says 6 tens and 1 one. The blocks really show 16.

3. Nina writes 81 for eighteen. Circle the misplaced digit and write eighteen correctly.

Step 1: say the number slowly — *eightteen* means 1 ten and 8 ones. Step 2: Nina's 8 is sitting in the tens place, so her number says 8 tens and 1 one. Circle that 8. Step 3: put the 1 ten first and the 8 ones second: $10 + 8$, so eighteen is 18.

4. Sam builds 1 ten and 3 ones; Ty builds 3 tens and 1 one. Write $<$ or $>$.

Step 1: build Sam's number — $10 + 3$ gives thirteen. Step 2: build Ty's number — $10 + 10 + 1$ gives thirty-one. Step 3: thirteen has only 1 ten and thirty-one has 3 tens, so Sam's number is the smaller one: $<$.

Teaching point: the same two digits, swapped, are not the same number.

5. Ana counts one full ten-frame and 7 more dots and writes 71. Say why 71 is wrong and write the true number.

Step 1: count the frames — one frame is full (1 ten) and the second frame has 7 dots (7 ones). Step 2: 71 would need 7 full ten-frames; there is only one full frame, so 71 is far too big. Step 3: 1 ten and 7 ones is $10 + 7$, so the dots show 17.

6. Jo swaps the digits of the number made from 1 ten and 9 ones. Compare Jo's swapped number with the number she meant.

Step 1: the number Jo meant is $10 + 9$, which is nineteen — 1 ten and 9 ones. Step 2: swapping the digits puts the 9 in the tens place, giving 9 tens and 1 one, which is ninety-one. Step 3: 9 tens is much more than 1 ten, so Jo's swapped number is the bigger one: >. The digit in the tens place of Jo's swapped number is the 9.